

ai-os-new

Contract Violation Re-Audit Report

Verification audit after bug fixes. Original: 23 findings. Remaining: 23 findings.

Severity	Count
CRITICAL	3
HIGH	7
MEDIUM	10
LOW	3
Total	23

CV-01	CRITICAL	Resume from Pause broken - session stuck forever
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Files: debate-session.ts, debate-engine.ts

Problem: VALID_TRANSITIONS['paused'] doesn't include 'queued'. resumeSession startsession tries paused to queued which is silently rejected. Session permanently stuck.

Fix: Add 'queued' to VALID_TRANSITIONS['paused']. Or transition to 'deliberating' first in resumeSession.

CV-02	CRITICAL	Mutable internal references leaked via getters
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Files: debate-session.ts, key-state-projection.ts, router-projection.ts, debate-room.ts

Problem: agentStates getter returns live internal Map. KeyStateProjection.getState() and RouterProjection.getState() return internal Maps. Callers can directly mutate.

Fix: Return defensive copies or ReadonlyMap wrappers.

CV-03	CRITICAL	CacheService.get() mutates on read
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Files: cache-service.ts

Problem: get() is read by name but mutates entry.hitCount++, deletes and re-inserts for LRU, and returns same mutated object reference.

Fix: Return a copy. Track hit count internally on the cache entry copy.

CV-04	HIGH	CircuitBreaker getState()/checkAndGetState() mutate on read
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Files: circuit-breaker.ts

Problem: Both call updateAndGetState() which transitions OPEN to HALF_OPEN. Monitoring calls can inadvertently trigger state transitions.

Fix: Split into peekState() (pure read) and getState() (with side-effect, document intent).

CV-05	HIGH	KeyLifecycle has no transition table - any state can jump anywhere
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Files: key-lifecycle.ts

Problem: transition() accepts any from-to pair. onError can push quarantined to degraded directly. onSuccess can go degraded to active skipping probation.

Fix: Define and enforce LIFECYCLE_TRANSITIONS table.

CV-06	HIGH	KeyHealth.checkHealth() mutates key object as side effect
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Files: key-health.ts

Problem: checkHealth() mutates key.latency, key.status, key.stats.avgLatency, calls saveKeys() and notify(). Diagnostic check corrupts state.

Fix: Return result without mutating key. Let caller decide whether to apply changes.

CV-07	HIGH	ConfigService.updateRouter() silently discards input
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Files: config-service.ts

Problem: Method accepts Partial<RouterConfigSection>, logs deprecation warning, completely discards input without throwing. Callers believe config applied.

Fix: Throw explicit error so callers know update was not applied.

CV-08	HIGH	VirtualKeyService.resolve() mutates returned object and triggers persist
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Files: virtual-key-service.ts

Problem: resolve() mutates vk.lastUsedAt on cached object, triggers debouncedPersist. Read operation triggers write. Returned object is same reference as cache.

Fix: Return defensive copy. Separate read from mutation.

CV-09	HIGH	toggleKeyStatus() can hide error/quarantined state
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Files: key-health.ts

Problem: Toggles between inactive/active but if status is error/quarantined/compromised, sets to inactive hiding the error state.

Fix: Only toggle between active and inactive. No-op for other statuses.

CV-
10

HIGH

DebateBudget accepts sessionId but never validates it

Files: debate-budget.ts

Problem: canProceed() and recordUsage() accept sessionId parameter that is never used - always operates on this._sessionId. Wrong ID modifies wrong budget.

Fix: Validate sessionId matches this._sessionId or remove parameter.

CV-
11

MEDIUM

HealthScoreService.getScore() mutates state on read

Files: health-score-service.ts

Problem: When cache stale, getScore() calls computeScore() which writes to this.scores and this.history. Polling loop grows history unbounded.

Fix: Return cached data. Separate recompute into explicit refresh() method.

CV-
12

MEDIUM

DebateSession.transition() silently swallows invalid transitions

Files: debate-session.ts

Problem: Invalid transitions produce only console.warn and return silently. Caller has no way to know transition was rejected.

Fix: Return boolean or throw on invalid transition.

CV-
13

MEDIUM

AgentService lifecycle has no transition validation

Files: agent-service.ts

Problem: Any AgentLifecycleState can transition to any other. Terminated to busy, initializing to paused possible.

Fix: Define valid transition map for agent lifecycle states.

CV-
14

MEDIUM

KeyVault.purgeKey() mutates external object via cast

Files: key-vault.ts

Problem: purgeKey(key) casts key as {key?:string} and sets key=". Bypasses TypeScript readonly. Corrupts caller's reference.

Fix: Return new object instead of mutating. Or document in-place mutation intent.

CV-
15

MEDIUM

MemoryRepository.computeId() collision-prone

Files: memory-repository.ts

Problem: Hash uses string lengths (content.length.toString(36)) not content. Different strings of same length produce same prefix.

Fix: Use proper content-based hash (SHA-256 or similar).

CV-
16

MEDIUM

VirtualKeyService.init() silently falls back to memory-only

Files: virtual-key-service.ts

Problem: If DB not ready, falls back to in-memory mode with console.warn. Caller can't detect persistence disabled. Keys lost on reload.

Fix: Expose persistenceEnabled property. Log warning more prominently.

CV-
17

MEDIUM

CognitiveService.evaluateAlternatives() always empty

Files: cognitive-service.ts

Problem: Returns [] always. executeAgentNode() always throws 'No viable execution alternatives'. makeDecision() unreachable.

Fix: Either implement or throw 'not implemented' immediately.

CV-
18

MEDIUM

HealthScoreService accesses ProviderTracker via unsafe cast

Files: health-score-service.ts

Problem: Uses (this as unknown as {providerTracker?:...}).providerTracker - never set. getScore() always returns null.

Fix: Make providerTracker a proper constructor dependency.

CV-19	MEDIUM	KeyHealth.getHealthUrl() silent fallback to OpenAI for unknown providers
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Files: `key-health.ts`

Problem: Unknown providers fall back to `api.openai.com/v1/models`. Custom provider health check reports false 'healthy' by checking OpenAI instead.

Fix: Throw for unknown providers. Or return error status.

CV-20	MEDIUM	TransactionContext.commit() partial failure handling
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Files: `transaction.ts`

Problem: On commit failure, `_committed=false` then `rollback()` called. Some persists may have succeeded. Transaction in half-processed state.

Fix: Set `_committed=true` only after ALL persists succeed. Then emit events.

CV-21	LOW	Cross-tab broadcastCompatibility creates bogus entries
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Files: `cross-tab-state.ts`

Problem: Creates `CircuitBreakerState/RateLimitState` with `provider:'unknown', remaining:0`. Stored in local maps and synced to other tabs.

Fix: Validate inputs before creating state entries.

CV-22	LOW	ResumableStream.resume() starts fresh
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Files: `resumable-stream.ts`

Problem: `resume()` sends full request body again without offset. Replays all content from beginning. Caller receives duplicate content.

Fix: Track last yielded index. Skip already-yielded chunks on resume.

CV-23	LOW	debate-session-persistence silently swallows all errors
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Files: `debate-session-persistence.ts`

Problem: Every function wraps body in `try/catch` with `console.warn` returning `null/[]`. Critical persistence failures silent.

Fix: Propagate critical errors. Only swallow truly transient/read errors.